

Review

Aquifer pollution by metal-antibiotic complexes: Origins, transport dynamics, and ecological impacts

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ABSTRACT

Aquifer pollution by metal-antibiotic complexes is a rising environmental and public health concern owing to their enhanced mobility and persistence in groundwater. The purpose of this review is to examine the origins, transport dynamics, and ecological impacts of complexes formed through interactions between metal ions and antibiotics in agricultural runoff, pharmaceutical effluents, and wastewater discharge. Metal-antibiotic complexes are more resistant to degradation and are more soluble than their components. This complicates the conventional water purification efforts. These complexes disrupt microbial ecosystems, facilitate the spread of antibiotic-resistance genes, and negatively affect aquatic organisms. The entry of pollutants into drinking water sources poses notable health risks, including chronic exposure to contaminants and the emergence of antibiotic-resistant pathogens. This review emphasizes both preventative and remedial strategies to mitigate these impacts. Preventative measures emphasize the regulation of antibiotic and metal use in agriculture and industry and promote green chemistry alternatives. Remediation approaches include advanced treatment technologies such as membrane filtration, oxidation, and bioremediation. Integrated management practices and ongoing monitoring were discussed to address this complex issue. To protect water quality and public health, metal-antibiotic complexes in aquifers require stringent regulatory measures, innovative treatment solutions, and heightened public awareness. This review highlights the importance of coordinated efforts to prevent and remediate the emerging pollution problem.

1. Introduction

1.1. Background on aquifer pollution

Underground water reservoirs, also known as aquifers, are essential sources of drinking water and irrigation (Laonamsai et al., 2023; Sarawat et al., 2023). Aquifer pollution occurs when harmful substances are introduced into these water sources either naturally or by humans. It is often difficult to identify the sources of this pollution because of its diversity and abundance (Pradhan et al., 2023). Most pollutants originate from agricultural runoff, industrial discharges, and leaking septic tanks. As a result of these activities, pollutants such as nitrates, chemicals, heavy metals, pathogens, organic pollutants (including antibiotics), pesticides, fertilizers, microplastics, pharmaceuticals, and personal care products are introduced into the environment (Khezami et al., 2024). Recently, attention has also been given to a unique type of contaminant known as antibiotic-metal complexes (AMCs), which are hybrids formed from the combination of antibiotics and metals (Yao

et al., 2020). Remediating contaminated aquifers presents significant challenges owing to the slow and complex movement of groundwater (Al-Hashimi et al., 2021), potentially leading to long-term environmental and public health risks, including waterborne diseases and ecosystem disruption.

1.2. Overview of metal-antibiotic complexes

The study of antibiotic-metal complexes (AMCs) has gained momentum over the past two decades, and these compounds have drawn increasing attention owing to their environmental persistence. While AMCs were originally formed in pharmaceutical laboratories, recent research has revealed that these complexes also form in natural environments, such as soil, air, and particularly in aquatic systems. Aquatic environments, rich in pollutants such as heavy metals, antibiotics, and organic matter, are hotspots for significant metal-antibiotic complexation (Jijie et al., 2023). The coexistence of these multiple pollutants can lead to simultaneous reactions, generating secondary metabolites such

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as reactive oxygen species (ROS), which further promote the formation of AMCs (Zhao et al., 2023). This discovery sparked global interest and prompted more careful consideration of the intrinsic chemistry of AMCs.

As emerging pollutants, AMCs exhibit significant differences in solubility, bioavailability, photolysis, and absorption compared with their parent antibiotic compounds (Zhang et al., 2012). The altered physicochemical properties of these residual antibiotics are primarily due to the formation of stable divalent and trivalent metal complexes, which are influenced by the structural characteristics of the metals and antibiotics (Cao et al., 2025; Chen et al., 2024a). Although low levels of AMCs may offer some benefits to animal health, excessive amounts can harm the environment, contributing to the development of antibiotic and metal resistance genes in bacteria (Lee et al., 2001). This accumulation poses risks to both humans and aquatic life through the food chain (Chahouri et al., 2023) and other pathways, highlighting the potential toxicity of aquifers contaminated with these co-contaminants. AMCs have also been observed to interact with genetic material, leading to genetic alterations or mutations, suggesting possible genotoxicity and adverse effects on organisms exposed to these environments over time (Wang et al., 2022). The modified biological and physical properties of AMCs often render them more harmful than their constituent components (Seiler and Berendonk, 2012).

Understanding the structure and physicochemical properties of AMCs is crucial for elucidating their mechanisms of action (Ramotowska et al., 2020). This involves the assessment of properties such as acidity, lipophilicity, electronic and spatial structures, stability, binding sites, and ligand donor atoms. Various analytical methods, including crystallography, computation, spectroscopy, and thermogravimetry, have been employed to study these complexes (Masoud et al., 2015). Crystallography provides structural details, computation predicts interactions, spectroscopy reveals the molecular composition, and thermogravimetry assesses thermal stability. By integrating these techniques, researchers have gradually developed a comprehensive understanding of AMCs. However, there are still many aspects that have not been explored, which underscores the need for advancements in instrumentation and analytical chemistry.

1.3. Importance of addressing this issue

The study of metal-antibiotic complexes is a burgeoning field that combines coordination chemistry with pharmacology. This study provides valuable insights into the effectiveness of these complexes in fighting microbes and their potential environmental impact. It is crucial to balance the therapeutic advantages of AMCs with their potential harmful effects, making further research essential for their future applications. Although most studies have focused on the individual detection, analysis, and effects of metals and antibiotics in the environment, there is a lack of exploration of the complex chemistry of AMCs in wastewater and sludge (Khurana et al., 2021a).

Wastewater effluents, often inadequately treated by traditional wastewater treatment plants (WWTPs), can contaminate nearby aquifers over time (Pradhan et al., 2023). The dilution of AMCs in aquifers makes their detection, characterization, and quantification challenging. Understanding the threat posed by AMCs requires a detailed analysis of their structural characteristics and the mechanisms underlying their toxicity. The presence of multiple co-existing chemicals complicates this assessment, as mixtures can exhibit additive or synergistic effects, a phenomenon sometimes referred to as the "something from nothing" effect (Gao et al., 2018). Although various methods for removing AMCs have been explored, comprehensive knowledge of their mechanisms and toxic effects is still limited (Wang et al., 2022). Thus, there is a crucial need to explore and analyze the simultaneous presence of antibiotics and heavy metal pollution in the environment. To date, very few studies have been conducted on the environmental toxicological effects of AMCs, and further studies on the environmental risks of AMCs are urgently required.

2. Origins of metal-antibiotic complexes

2.1. Sources of antibiotics in aquifers

Antibiotics have emerged as a significant environmental hazard, second only to metal pollutants, and pose a major challenge for environmental safety authorities. These "miracle drugs," widely used since their introduction in 1941 to treat bacterial infections, have seen a surge in usage, particularly post-World War II, impacting human, animal, and plant health (Saikia and Chetia, 2024). Antibiotics are classified into various groups as chemotherapeutic agents and antimicrobials, such as tetracyclines, nitrofurantoin, metronidazole, penicillin, quinolones, cephalosporins, aminoglycosides, macrolides, clindamycin, sulfonamides, trimethoprim, and tinidazole, which target specific bacteria and parasites. The growing use, misuse, and abuse of antibiotics, especially in developing countries, has led to their classification as emerging contaminants in various environmental compartments, including aquifers. The potentially harmful effects of antibiotics on living organisms, including animals and plants, have been extensively studied, making antibiotic toxicity a global concern (Gros et al., 2021). In aquatic environments, antibiotic residues typically range from nanograms per liter (ng/L) to micrograms per liter ($\mu\text{g/L}$) (McCorquodale-Bauer et al., 2023), whereas in solid matrices, the concentrations range from nanograms to micrograms per gram (Liu et al., 2023b). Major antibiotic classes such as macrolides (MAs), fluoroquinolones (FQs), tetracyclines (TCs), and sulfonamides (SAs) dominate environmental contamination, disrupting vital ecosystem functions such as nutrient cycling and decomposition by affecting microbiota (Gao et al., 2023).

The origin of these contaminants can be categorized into point and nonpoint sources. Point sources refer to specific locations where pollutants are directly released into water bodies, such as wastewater treatment plants (WWTPs), industrial facilities, hospitals, urban drainage systems, and commercial establishments (Addis et al., 2023). WWTPs are particularly significant contributors, as they often struggle to fully remove organic pollutants, including antibiotics, even with advanced treatment methods such as bioreactors, advanced oxidation processes, and nanotechnologies (Batt et al., 2006). Consequently, substantial levels of antibiotic residues are frequently found in WWTP effluents and sludge. Improper disposal of household antibiotics, pharmaceutical industry waste, and discharges from healthcare facilities further exacerbates the contamination of water systems (Pang et al., 2024). Nonpoint sources of antibiotic contamination are more diffused and lack specific discharge points making them challenging to manage. Agricultural runoff, especially from livestock farming and aquaculture, is a major nonpoint source of runoff. Regions with concentrated livestock farming are particularly vulnerable because animal manure containing high levels of antibiotics can degrade groundwater quality and potentially reach aquifers (Gros et al., 2023).

2.2. Sources of metal contaminants

Heavy metal pollution in the environment is a significant concern and is driven by both natural processes and human activities. Industrialization, urbanization, and agricultural practices, such as the use of fertilizers and pesticides, have contributed to the release of toxic metals such as antimony, arsenic, cadmium, lead, and mercury into aquatic ecosystems (Hama Aziz et al., 2023). These metals, whether introduced through waste disposal, mining, electroplating, smelting, or natural phenomena, such as forest fires, weathering, and volcanic eruptions, are persistent pollutants that are not easily degraded. They can exist in soluble or suspended forms, settle at the bottom of water bodies, or be absorbed by organisms, leading to long-term environmental and health impacts (Heidari and Kordbacheh, 2023).

Human activities significantly contribute to the pollution of water bodies. For instance, mining activities produce large amounts of metal containing waste, which are often released into the environment

without proper management. This leads to the accumulation of toxic metals in water bodies (Kumari and Bhattacharya, 2023; Ezekiel et al., 2023). Urban and industrial areas usually have high concentrations of metals such as silver, arsenic, cadmium, copper, chromium, mercury, nickel, lead, and zinc. This is mainly due to emissions from vehicles, coal burning, and improper handling of electronic wastes (Isinkaralar et al., 2023). Conversely, natural water sources generally contain essential nutrients, such as potassium, calcium, magnesium, iron, and manganese. However, imbalances in these nutrient levels can also result in toxic effects on aquatic life (Kolarova and Napiórkowski, 2021).

Contamination of aquifers and other water bodies with metals poses serious risks to human health and the environment. These pollutants tend to accumulate in inshore waters and estuaries, posing a threat to aquatic life and causing mass mortality among species (Ismanto et al., 2023; Chahouri et al., 2023). Fish, as key components of the aquatic food chain, often accumulate high levels of these metals, leading to long-term toxicity that affects not only aquatic organisms but also higher trophic levels, including humans (Gokul et al., 2023). The processes of bioaccumulation and biomagnification exacerbate the impact of these pollutants across different trophic levels, making metal contamination a critical global issue (Saidon et al., 2024). Fig. 1 illustrates the sources of metal antibiotic contaminants in aquatic environments and their impact on aquifers.

2.3. Formation of metal-antibiotic complexes

Transition metals, such as Cu, Fe, Zn, Ni, and Co are known for their

ability to form stable complexes with organic drugs and antibiotics. This is because of their diverse reactivity and coordination properties, such as variable oxidation states, chelating ability, Lewis acid behavior, and redox properties (Bjerrum, 1950). These metals can be likened to soldiers, with robust ligands acting as armor that protects them from harm (Xu et al., 2022). The concept of metal complexation in antibiotics has been employed in the pharmaceutical field since the early 1940s and has significantly influenced drug development. Metal-based medications derived from these complexes show promising potential for treating severe illnesses such as cancer, malaria, and neurodegenerative disorders (Elattar et al., 2024). The growing threat of antibiotic resistance has created an urgent need for new antimicrobial compounds (Frei et al., 2020; Sharma et al., 2022).

Research in this area has led to the development of antibiotic-metal complexes with enhanced efficacy and reduced toxicity compared to traditional antibiotics. These complexes have garnered attention because of their ability to improve antibiotic activity, overcome bacterial resistance, increase drug bioavailability, and reduce dosage requirements (A. El-Habeeb et al., 2024). The variety of available metal cores and ligands offers numerous opportunities for modifying and enhancing existing drugs, paving the way for new antimicrobial treatments (Regiel-Futyrta et al., 2017).

Metal ions such as silver, copper, zinc, and gold are frequently used in these complexes because of their inherent antimicrobial properties (Turner, 2024). The coordination of these ions with antibiotic ligands can produce synergistic effects, where the combined activity is greater than that of individual components (Chen et al., 2024a). Notable

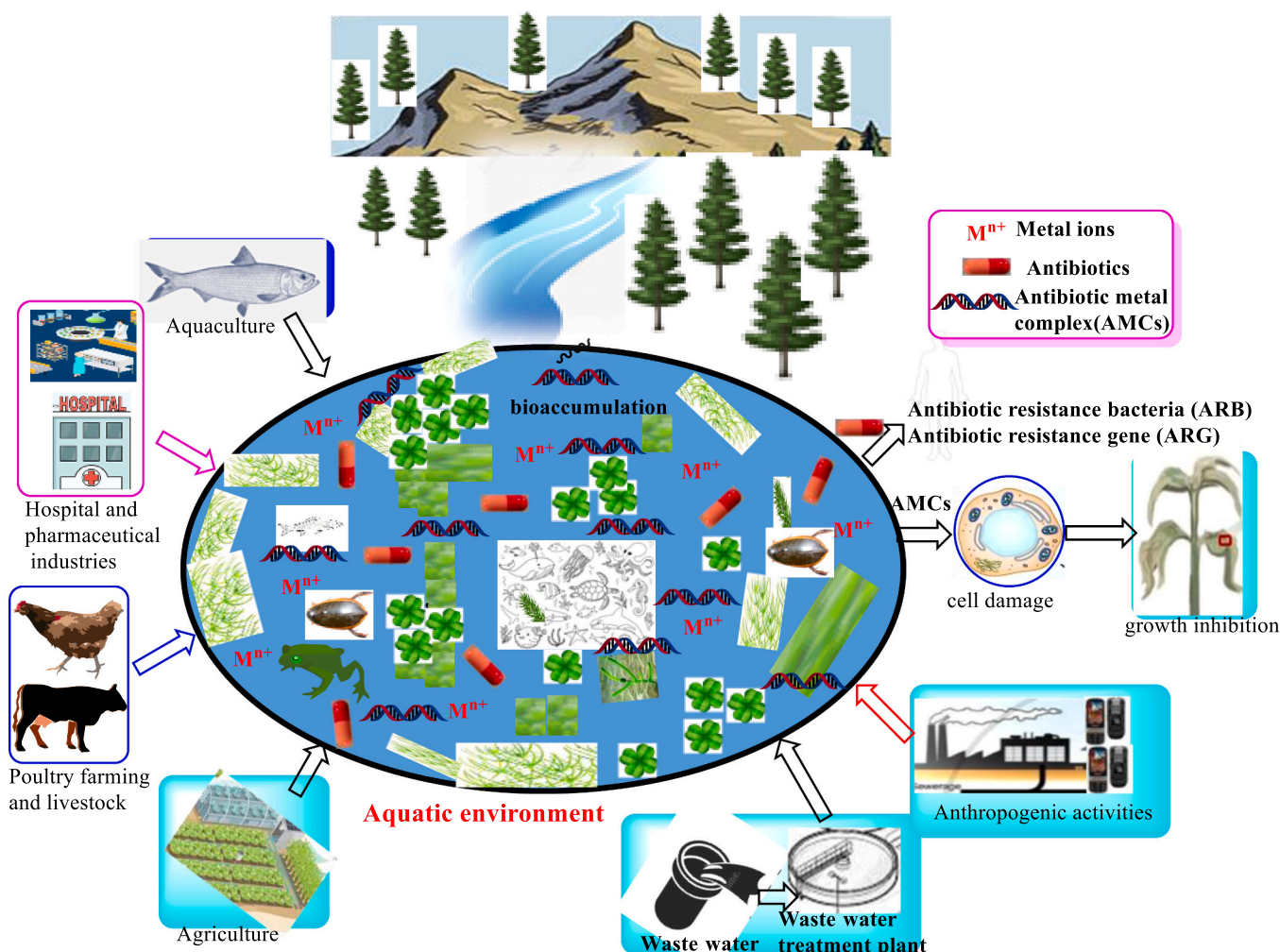


Fig. 1. Sources of antibiotics and metals contaminating aquifers.

examples include metalloantibiotics, vancomycin-metal complexes, copper(II) complexes with quinolone antibiotics, and silver sulphadiazines. While the effects of these complexes on targeted species have been well-studied, there is insufficient research on their potential toxicity and stability issues in non-targeted species. In animal husbandry, heavy metal salts of zinc and copper, along with antibiotics such as sulfonamides, fluoroquinolones, and tetracyclines, are commonly added to animal feeds to boost health and productivity (Pulicharla et al., 2017; Yao et al., 2020). However, these substances are not fully absorbed by livestock and are often excreted, leading to environmental contamination when manure is used as a fertilizer (Wang et al., 2021). Wastewater containing a mixture of antibiotics and heavy metals, often from sources such as pharmaceuticals, hospitals, and agricultural fields, is discharged into water bodies because of the inefficiency of traditional wastewater treatment plants (WWTPs). The coexistence of these metals and antibiotics leads to the formation of AMCs in water bodies.

Metal-antibiotic complexation can occur through different mechanisms. Many commonly used antibiotics contain nitrogen-containing heterocycles with a six-electron cyclic aromatic system that readily undergo electrophilic substitution reactions with metal ions (Wang et al., 2022). Functional groups like -OH, -COOH, -NH₂, and -SH facilitate complexation with metal ions via coordination bonds, where partially filled d-orbitals of metal ions accept electron pairs from these donor groups (Paul et al., 2014). Fig. 2 illustrates the electronic pathway and mechanism of the AMCs formation. The chemical structures of the AMCs with metal coordination sites are shown in Fig. 3.

According to ligand field theory, the electronic structure of metal ions and the spatial arrangement of ligands determine the stability, reactivity, and geometry of the complexes. Metal ions with partially filled d-orbitals (e.g., Fe²⁺, Cu²⁺, and Ni²⁺) exhibit higher affinity for electron-rich antibiotic functional groups compared to those with half-filled (e.g., Mn²⁺) or fully filled (e.g., Zn²⁺) d-orbitals. This increased stability arises from interelectronic interactions between ligand and metal d-orbitals, whereas ion-dipole interactions dominate in the latter. The higher Fe²⁺ concentration in poultry litter compared to Mn²⁺ and Zn²⁺ serves as strong evidence for this, as Fe²⁺ binds more strongly with antibiotics such as doxycycline and ofloxacin (Khurana et al., 2021a). Metal ions favor specific coordination geometries (e.g., octahedral, tetrahedral, and square planar) based on their electronic configurations. Antibiotics with multiple binding sites may face steric hindrance, limiting their utilization, whereas chelate formation enhances complex stability when binding site orientation aligns with the preferred geometry of the metal ion. Additionally, the heterocyclic moieties and aromatic rings in antibiotics allow π - π interactions and coordination with metals (Puri et al., 2021). Environmental pH plays a crucial role in the ionization state of both antibiotics and metals, affecting their ability to form complexes. For instance, at a low pH, metals are more likely to exist as free ions, enhancing complexation with antibiotics. The likelihood of complex formation also increases with higher concentrations of either

antibiotics or metal ions. Microplastics can provide adsorption sites for both metals and antibiotics, potentially facilitating their interactions (Lv et al., 2023). Competing cations (e.g., Na⁺, K⁺) or anions (e.g., Cl⁻, SO₄²⁻) can interfere with antibiotic-metal complexation by binding to the antibiotic or metal. The stability of these complexes typically increases with the nuclear charge of the metal ions. Consequently, heavy metal ions, which have higher nuclear charges, form more stable and persistent complexes, posing a greater environmental threat (Cuprys et al., 2018). Although complexation can reduce the potential harm of the constituent metal ions and antibiotics, it also increases their persistence in the environment. The experimental data indicate that AMCs exhibit higher persistence at low pH (acidic to neutral) and low temperatures. AMCs undergo photolysis, hydrolysis, and microbial degradation at high pH and elevated temperatures, leading to reduced persistence. The ionization of an antibiotic's functional groups depends on the pH of the environmental medium, leading to conformational changes that allow the antibiotic to act as a monodentate, bidentate, or bridging ligand. The order of preference of the donor sites for metal complexation also varies with pH. The stoichiometry of these complexes is pH-dependent (Carlotti et al., 2012). For instance, in tetracycline-divalent metal ion complexation, the antibiotic remains uncomplexed in strongly acidic media regardless of the metal ion concentration, while 1:1 and 2:1 (M:L) stoichiometries are observed at higher pH. Similarly, for ciprofloxacin-metal complexation, 1:1 ratios are favored at acidic pH, whereas 1:2 ratios dominate at higher pH. As the temperature increases, the stability constant of AMCs decreases, thereby reducing their persistence owing to an increased likelihood of dissociation caused by natural processes such as hydrolysis and photolysis. Table 1 shows the relationship between the stability constants and temperature for various AMCs and Table 2 reports the different classes of antibiotics complexed with metals and their useful applications.

3. Transport dynamics

3.1. Mobility in aquifer systems

The adsorption of AMCs onto soil particles, especially clay minerals and natural organic matter, reduces their mobility (Nkoh et al., 2024). This process is influenced by factors, such as soil composition, pH, and the presence of competing ions or organic substances (Jia et al., 2008). High adsorption can stabilize the complex within a localized area, whereas low adsorption may enhance mobility and increase the likelihood of dissociation. The solubility of AMCs in water plays a crucial role in their transport through aquifers; those with higher solubility are more mobile and can travel further within the groundwater. The chemical stability of AMCs, governed by environmental conditions, such as pH and redox potential, determines their persistence and potential transformation into other compounds.

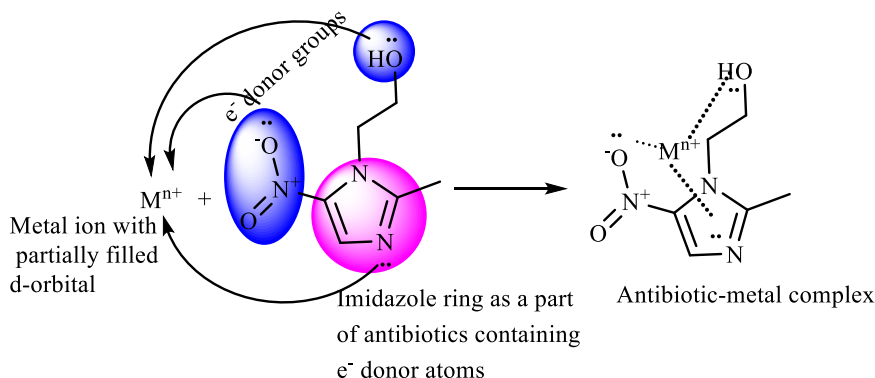


Figure 2. Graphical representation of mechanism of antibiotic-metal complexation.

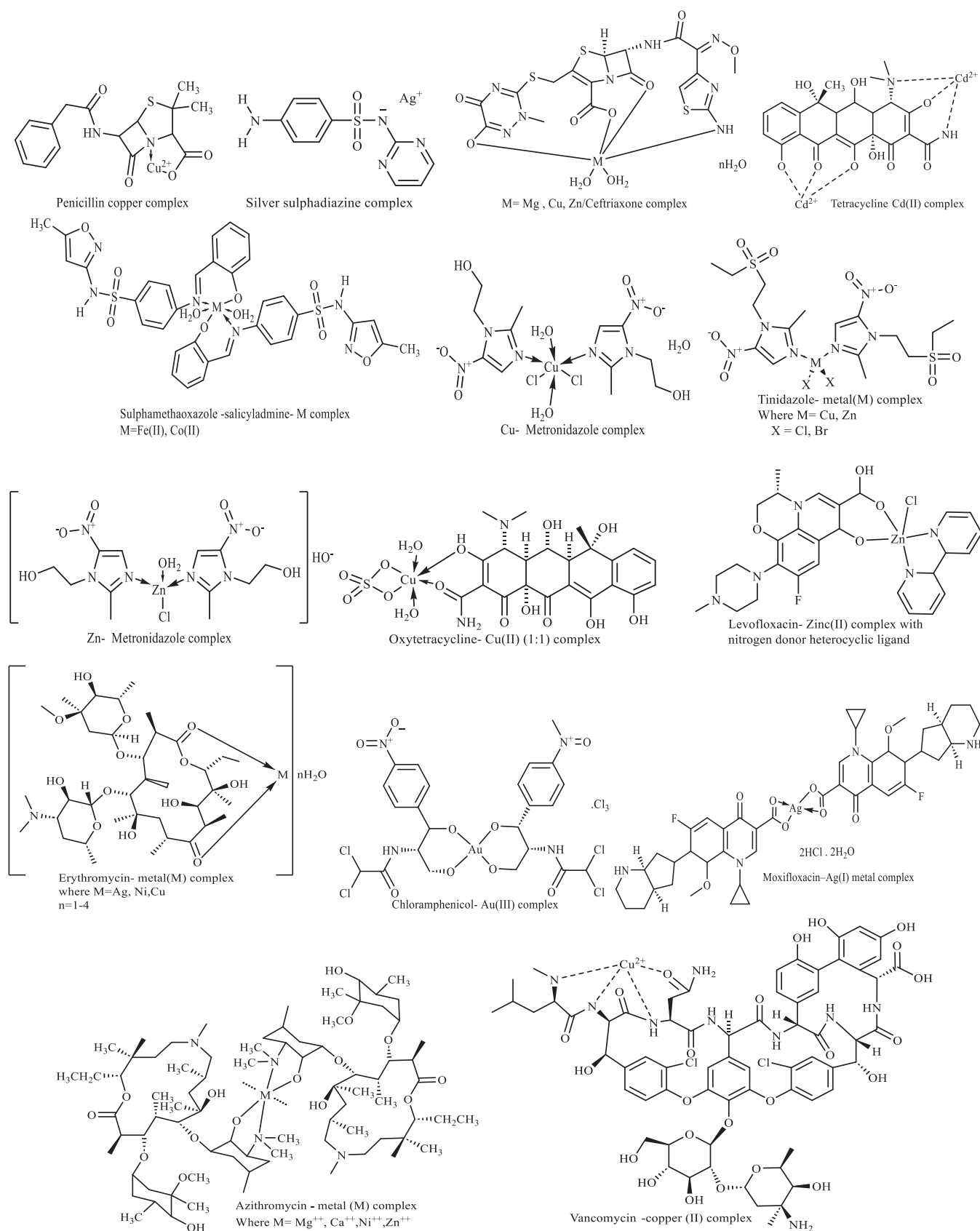


Fig. 3. Chemical Structures of some of AMCs discussed in the paper.

Table 1
Dependence of stability constant of AMCs with temperature.

Antibiotics	Metal ions	Temperature (K)	Stability constant	Reference
Levofloxacin	Mn(II)	308	9.506	(Chouhan et al., 2022)
		313	9.205	
		303	10.449	
Amoxicillin	Mn(II)	308	10.138	
		313	9.84	
Amoxicillin	Cu(II)	298	1.3±0.7	
		288	2.36	
Ampicillin	Mn(II)	298	2.37	
		310	2.09	
Ampicillin Trihydrate; D [-]-α-aminobenzyl penicillin; (ABP)	Co(II)	298	2.68	(Gupta, 2014)
		308	2.66	
	Ni(II)	318	2.65	
		298	2.67	
		308	2.66	
	Cu(II)	318	2.64	
		298	2.77	
		308	2.66	
		318	2.65	

3.2. Factors influencing transport

The fate and transport of AMCs in aquifer systems are influenced by various factors including soil pH, hydrological conditions, and the presence of competing ions. Soil pH plays a crucial role in determining the charge and solubility of AMCs. In acidic environments, metal ions tend to be more mobile, whereas under alkaline conditions, they are more likely to precipitate or adsorb onto the soil particles. Additionally, the soil composition, such as organic matter and mineral content, significantly affects the adsorption and mobility of these complexes. Hydrological factors such as groundwater flow velocity, direction, and hydraulic conductivity also play a key role in the transport of AMCs. Faster flow rates can facilitate the spread of AMCs across a broader area, whereas slower rates may restrict their movement. The existence of preferred flow paths such as fractures or zones with higher permeability can further influence the transport dynamics of AMCs. The presence of competing ions in groundwater, such as calcium, magnesium, and sodium, can disrupt the stability of AMCs by either replacing metal ions in the complexes or occupying available adsorption sites. This interaction directly affects the mobility and transport behavior of AMCs within the aquifer system.

A case study on the Yitong River in China highlighted significant seasonal and spatial variations in antibiotic and heavy metal pollution. Major contaminants include antibiotics such as ofloxacin (OFL), norfloxacin (NOR), sulfamethoxazole (SMZ), and tetracyclines (TC), along with metals such as cadmium, lead, and copper. These pollutants originate from untreated domestic and industrial wastewater, form stable antibiotic-metal complexes (AMCs). During heavy rainfall, active transport of AMCs was observed. Ecological risk assessments indicate high threats to primary aquatic producers, with fish and invertebrates experiencing impaired growth and reproduction. The river is also a hotspot for antibiotic resistance genes (ARGs)(Zhao et al., 2023).

4. Potential degradation pathways of Antibiotic-metal complexes (AMCs)

Photodegradation, biodegradation, and chemical degradation are the key pathways for the break down AMCs in the environment. Photodegradation, in particular, uses UV light to break metal-ligand bonds or alter the structure of antibiotics. This method is the most effective for mitigating pollution in shallow waters, where light penetration is the greatest. Bioremediation utilizes specific bacteria and fungi that metabolize antibiotics within a complex to facilitate degradation. Chemical degradation of the complex can be achieved by altering environmental conditions such as pH and redox potential, making the

antibiotic within the complex more susceptible to decomposition.

5. Ecological impacts

5.1. Ecological and biological impact of AMCs on aquatic ecosystems

Antibiotic-metal complexes can disrupt the composition and function of aquatic microbial communities by selectively inhibiting or promoting the growth of specific bacteria, algae, and fungi. This disruption impacts essential processes, such as nutrient cycling, including nitrogen fixation, phosphorus cycling, and organic pollutant degradation. Consequently microbial diversity is reduced, leading to imbalances in aquatic ecosystems. According to the chelation theory, AMCs enhance antimicrobial activity by allowing metal ions to share their positive charge with ligand donor groups, thereby decreasing ion polarity. This interaction causes overlap of the ligand and cation orbitals, leading to π -electron delocalization within the chelate ring, which increases the lipophilicity of the complex. Consequently, AMCs can easily pass through lipid membranes via passive diffusion, a process driven by a concentration gradient that does not require energy (Qian et al., 2024). In addition, active transport mechanisms may involve specific membrane proteins that recognize the metal complex and facilitate its entry into the cell via ATP. The toxicity of AMCs often surpasses that of the parent antibiotics (Zhang et al., 2012; Khurana et al., 2021b), affecting critical aquatic species such as bacteria and algae. For instance, amoxicillin-metal complexes exhibit greater antibacterial efficacy against *E. coli* than free amoxicillin by protecting the antibiotic from oxidation (Hrioua et al., 2021). Traditional toxicity assessments have frequently overlooked the combined effects of pollutants in natural environments (Wang et al., 2021). Algal communities, which serve as primary producers, are particularly sensitive to AMCs and can be used as biological indicators of toxicity (Liu et al., 2018). In eukaryotic algae, AMCs may enter cells through endocytosis, where they interact with cellular targets, inhibit growth, disrupt photosynthesis, and induce oxidative stress, thereby affecting both protein biosynthesis and carbohydrate uptake (Du et al., 2018). Toxicity tests on algae, such as *Myriophyllum aquaticum* and *Isochrysis galbana*, revealed that while antibiotics alone inhibited growth, the addition of metal ions significantly enhanced this inhibition. This enhanced toxicity is due to the oxidative stress and lipid damage caused by the unique chemical properties of AMCs, which increase their toxicity (Wang et al., 2022). Figs. 4 and 5 illustrate the pathways through which AMCs penetrate bacterial and algal cells as well as their subsequent biological effects.

The interactions between specific antibiotics and metal ions result in varying toxic effects depending on the nature of the antibiotic, the metal ion, the chelation effect, and the charge of complex. For instance, chlortetracycline (CTC) complexed with copper or cadmium exhibited antagonistic toxicity, whereas additive toxicity was observed between florfenicol (FLO) and the same metals because of the absence of interactions (You et al., 2022). Cationic complexes are generally the most toxic, followed by zwitterionic and anionic complexes, with dinuclear complexes being more potent than mononuclear complexes (Cuprys et al., 2018). The toxicity of AMCs can be particularly high for certain combinations, such as cadmium with tetracyclines or quinolones (Pulicharla et al., 2017). These combinations have been shown to inhibit phospholipase A2 activity and alter the microbial structure, enzyme activity, and substrate utilization, thereby affecting biodegradation processes (Khurana et al., 2021b). The differing EC50 values observed for various antibiotics combined with different metal ions highlight varying levels of toxicity toward the same microorganism, such as *Vibrio fischeri*. Even AMCs composed of similar classes of antibiotics and metals can exhibit different toxicity levels, indicating the importance of the careful selection of antibiotics because of their inherent toxic effects. The varying sensitivities of microorganisms and the potential environmental impacts of AMCs emphasize the need for further research particularly given their synergistic toxic effects and potential to disrupt

Table 2

Different classes of antibiotics with metal complexation and their biological profiles.

SN	Class	Example	Metal ions for complex formation	Useful applications	References
1	Penicillin	Amoxicillin	Cu(II) Cd(II)	Antibacterial	(Chaudhary and Mishra, 2018)
		Penicillin	Cu(II)	Treat bacterial infections like strep throat, syphilis	(Orta-Rivera et al., 2023)
		Cephalosporin	Sn(II)	Anti-bacterial, Used for respiratory tract infections, urinary tract infections	(Anaconda et al., 2012)
2	Sulphonamide	Ceftriaxone	Cu(II), Zn(II), Co(II)	Effective against bacterial meningitis, sepsis	(Anaconda et al., 2021)
		Sulfamethoxazole	Cu(II), Cd(II), Pb(II)	Used for urinary tract infections, bronchitis	(Rostamizadeh et al., 2019)
		Sulfadiazine	Zn(II)		
		Sulfisoxazole	Ni(II), Co(II)	Effective against bacterial meningitis, burns	(Rocha et al., 2019)
3	Tetracycline	Sulfamethizole	Zn(II), Cu(II)	Treats urinary tract infections, ear infections	(Kremer et al., 2006)
		Tetracycline	Zn(II), Cu(II)	Used for urinary tract infections	(Alalawy et al., 2022)
		Doxycycline	Cu(II) Ca(II), Mg(II), Fe(III), Fe (II)	Treats acne, cholera, malaria	(Zhang et al., 2012)
			Fe(III)	Effective against respiratory infections, Lyme disease	(Obaleye and Abosede, 2019)
		Minocycline	Ca(II), Mg(II), Fe(III)	Anti-inflammatory	(Zhang et al., 2019)
4	Quinolones	Tigecycline	Ca(II), Mg(II), Fe(III)	Anti-bacterial	(Arias et al., 2016)
				Anti-inflammatory	
		Ciprofloxacin	Cu(II), Ni(II) Co(II)	Anti-bacterial, anti-fungal, anti-oxidant	(Abd El-Lateef et al., 2023)
		Norfloxacin	Pd(II)	Anti-bacterial	(Shobana et al., 2024)
		Levofloxacin	Co(II)	Anti-bacterial, anti-oxidant	(Singh et al., 2023)
5	Macrolides	Moxifloxacin	Mg(II), Ca(II), Al(III)	Treats bacterial infections, anti-fungal	(Akhtar and Imran, 2024)
		Azithromycin	Zn(II), Cu(II), Co(II)	Treats bacterial infections such as bronchitis, pneumonia	(Sultana, 2014)
			Fe(III) Ni(II)		
		Erythromycin	Ni(II)	Anti-bacterial	(El-Shabrawy et al., 2024)
		Clarithromycin	Zn(II), Mg(II)	Anti-bacterial	(Nechifor et al., 2024)
6	Nonsteroidal	Roxithromycin	Fe(II) Fe(III)	Anti-microbial	(El-Rjoob et al., 2012)
		Ibuprofen	Cu(II), Mg(II), Zn(II)	Pain relief, anti-inflammatory	(Wang et al., 2022)
		Diclofenac	Zn(II), Cu(II), Ni(II)	Treats pain, inflammation	(Focşa, 2023)
7	Glycopeptides	Vancomycin	Ga(III) Fe(III)	Treats severe bacterial infections	(Wang et al., 2023)
		Teicoplanin	Co(III)	Anti-bacterial	
		Dalbavancin	Cu(II), Zn(II)	Anti-bacterial	(Zarkan et al., 2017)
		Oritavancin	Cu(II), Zn(II)	Anti-bacterial	(Zarkan et al., 2017)
8	Aminoglycosides	Gentamicin	Cu(II), Zn(II), Co(II)	Anti-bacterial	(Wang et al., 2022)
		Amikacin	Cu(II), Zn(II), Co(II)	Anti-microbial	(Abousaty et al., 2024)
		Tobramycin	Cu(II)	Anti-inflammatory	http://www.chemrj.org/ (Gziut et al., 2013)
		Neomycin	Cu(II), Zn(II), Co(III) Pd(II)	Anti-bacterial	(Al-Fakeh et al., 2024)
		Streptomycin	Pt(II), Au(I) Ag(I)	Anti-bacterial	(Ramasami et al., 2024)
9	Carbapenems	Imipenem	Co(II), Cu(II), Ni(II), and Zn(II)	Anti-bacterial	(Abousaty et al., 2024)
		Meropenem	Cd(II) Ag(I) Zn(II)	Anti-bacterial	(Umekar et al., 2021)
		Ertapenem	Cu(II)	Anti-bacterial	(Djoko et al., 2018)
10	Polypeptides	Bacitracin	Cu(II), Zn(II), Mn(II)	Anti-bacterial	(Ming and Epperson, 2002)
		Polymyxin B	Cu(II)	Anti-bacterial	(Gan et al., 2022)
		Colistin	Ga(I)	Anti-microbial	(Guo et al., 2023)
11	Lincosamides	Clindamycin	Mn(II)	Anti-bacterial	(Liu et al., 2023a)
		Lincomycin	AgNPs	Anti-bacterial	(Abdul-Jabbar et al., 2022)
12	Oxazolidinones	Linezolid	Ag(I) Au(I)	Antibacterial: multi-drug resistant Gram-positive infections	(Evans and Kavanagh, 2021)
13	Nitroimidazoles	Tedizolid	¹⁷⁷ Lu	Antibacterial: acute bacterial infections	(Karpuz et al., 2022)
		Metronidazole	Cu(II), Zn(II), Fe(III)	Antibacterial, Antiprotozoa	(Carnamucio et al., 2024)
		Tinidazole	Cu(II), Zn(II), Co(II) Ni(II)	Anthelmintic	(Alfaro-Fuentes et al., 2017)
14	Rifamycins	Rifampicin	Fe(III)	Anti-tumor	(Saad et al., 2020)
15	Nitrofurans	Rifabutin	Cu(II) Ca(II)	Antibacterial	(Wang et al., 2007)
		Nitrofurantoin	Cu(II)	Antibacterial	(Saad et al., 2020)
16	Phenicol	Furazolidone	Cd(II), Ni(II), Zn(II), Cu(II)	Antibacterial	(Wang et al., 2007)
		Chloramphenicol	Ca(II) Fe(III) Au(III)	Anti-cancer	(Al-Khodir and Refat, 2016)
17	Monobactams	Thiamphenicol	Zn(II), Cu(II), Ni(II), Co(II) and Fe(II)	Anti-microbial	(Sharfalddin et al., 2021)
				Anti-cancer	
18	Fosfomycin	Aztreonam	Zn(II)	Anti-microbial	(Díaz et al., 2004)
19	Streptogramins	Fosfomycin	Ni(II) Zn(II) Cu(II)	Anti-microbial	(Thompson et al., 2013)
20	Cyclic Lipopeptides	Quinupristin	Zn(II)	Anti-microbial	(Lipka et al., 2008)
21	Pleuromutilins	Daptomycin	Ca(II), Mg(II)	Anti-bacterial	(Ho et al., 2008)
22	Polyene	Retapamulin	Pb(II)	Anti-bacterial	(Tang et al., 2012)
23	Antifungals	Amphotericin B	Cu(II)	Antifungal: systemic fungal infections	(Chudzik et al., 2013)
		Nystatin	Zn(II)	Antifungal: Candida infections	(Andrejević et al., 2020)
23	Ionophores	Monensin	Co(II) Mn(II)	Antibacterial, Antiparasitic	(Dorkov et al., 2008)
		Salinomycin	Fe(III)	Antibacterial, Antiparasitic	(Petkov et al., 2024)

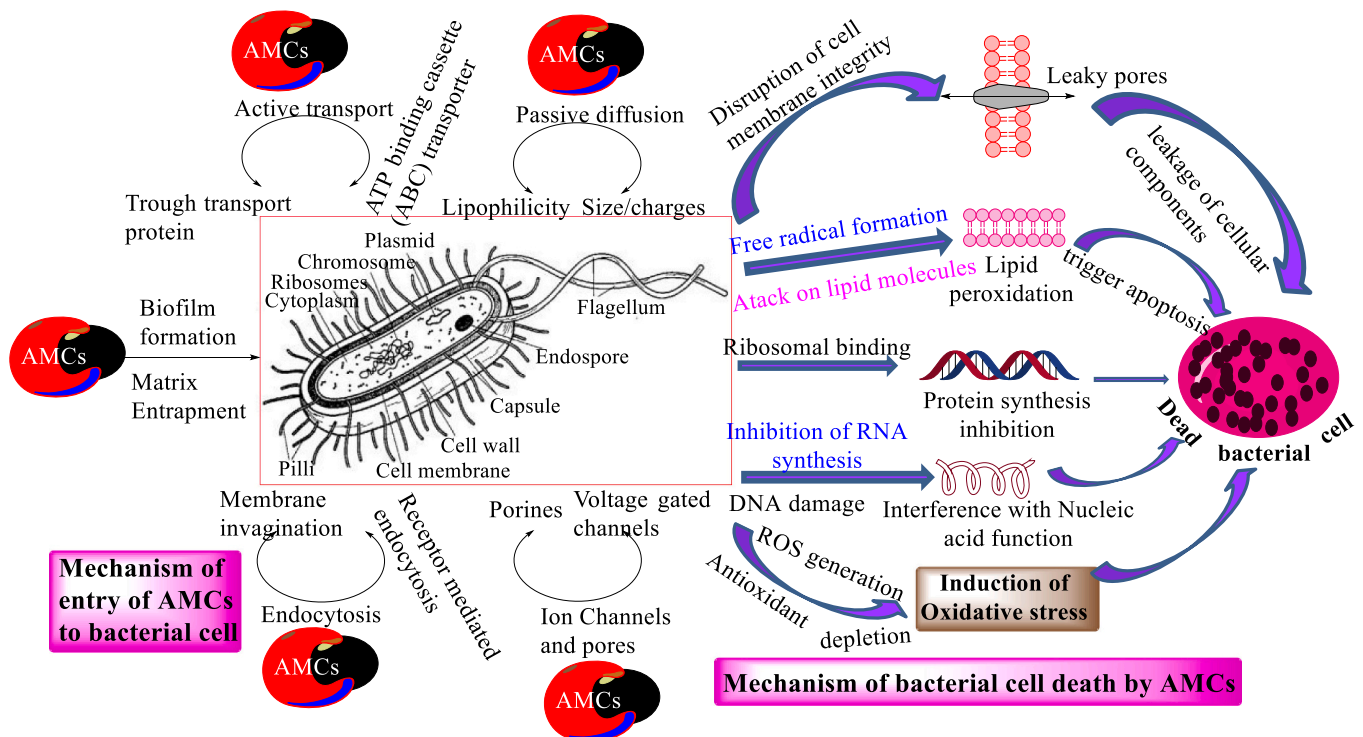


Fig. 4. Mechanism of entry and effect of AMCs in bacterial cell.

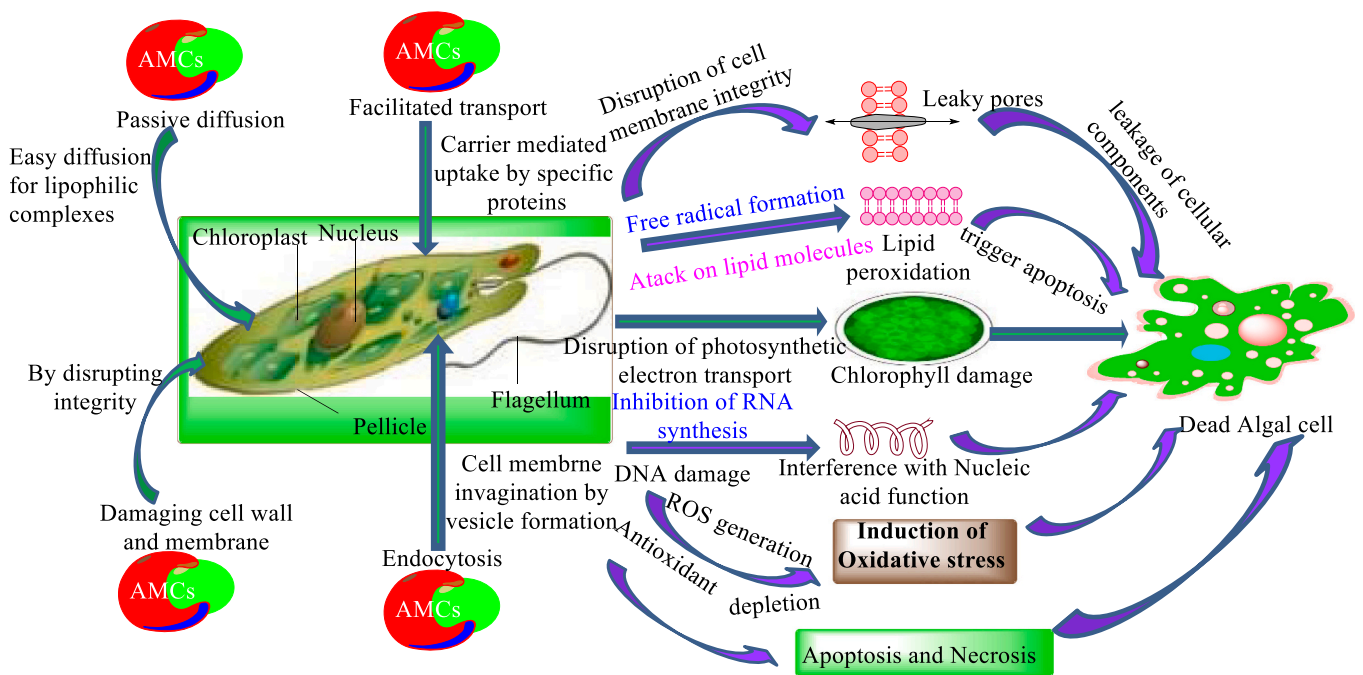


Fig. 5. Mechanism of entry and effect of AMCs in algal cell.

microbial ecosystems. Table 3 summarizes the interaction effects of various AMCs on different microorganisms, highlighting their potential synergistic, antagonistic, or additive effects.

5.2. Bioaccumulation and public health risks of antibiotic-metal complexes

The bioavailability of AMCs in aquatic systems is influenced by their solubility, chemical speciation, pH, temperature, and interactions with

natural organic and inorganic matter. Their binding to humic substances and sediments significantly affects their uptake by the organisms. Moreover, their fast absorption relative to slower metabolic and excretion rates results in bioaccumulation within aquatic organisms. Through food chain, bioaccumulated AMCs transfer from primary producers to upper trophic level consumers, leading to biomagnifications and poses significant ecological and health risks (Chouhan et al., 2022). AMCs can disrupt natural antibiotic degradation processes, such as photo-degradation leading to increased bioaccumulation (Chen et al., 2024b).

Table 3

Effect of antibiotic-metal complexes on various test organisms.

Antibiotic Metal Complexes (AMCs)	Effect Type	Test Organisms	Reference
Metronidazole-Cu(II)	Synergistic	<i>Escherichia coli</i> , <i>Candida albicans</i>	(Vianez Peregrino et al., 2021)
Sulphamethoxazole-Ag (I)	Synergistic	<i>Mycobacterium abscessus</i>	(Agertt et al., 2016)
Sulphadiazine-Zn	Synergistic	<i>Escherichia coli</i> and <i>Staphylococcus aureus</i>	(Anacona et al., 2015)
Tetracycline-Mg(II)	Additive	<i>Vibrio cholerae</i>	(Ray et al., 2017)
Levofloxacin-Zn(II)	Synergistic	<i>Escherichia coli</i>	(Galani et al., 2014)
Ciprofloxacin-Fe(II)	Antagonistic	<i>Pseudomonas aeruginosa</i>	(Wong et al., 2000)
Tetracycline-Cu(II)	Antagonistic	<i>Vibrio fischeri</i>	(Tong et al., 2015)
Tetracycline-Cd(II)	Synergistic	<i>Vibrio fischeri</i>	(Tong et al., 2015)
Ampicillin-Fe complex	Synergistic	<i>Staphylococcus aureus</i>	(Kosaristanova et al., 2023)
Chlorotetracycline-Cu (II) /Cd(II)	Antagonistic	<i>Synechocystis</i> sp.	(You et al., 2022)
Florfenicol- cu(II)/Cd (II)	Additive	<i>Synechocystis</i> sp.	(You et al., 2022)

For example, research on zebrafish exposed to ciprofloxacin-lead complexes revealed higher accumulation in tissues compared to ciprofloxacin alone, indicating enhanced bioavailability of AMCs in aquatic environments (Jijie et al., 2023). This raises concerns about human exposure through the consumption of contaminated fish as well as potential health risks, including gastrointestinal issues, organ damage, and carcinogenic effects. The presence of AMCs in water sources can render water unsafe for consumption, further threatening aquatic ecosystems by disrupting biodiversity and altering species composition. This disruption may lead to cascading effects throughout the food chain, impacting higher trophic levels such as fish and mammals. While terrestrial invertebrates possess detoxification mechanisms to mitigate the toxic effects of AMCs (Ahearn et al., 2004), mammals, including humans, are more vulnerable to the bioaccumulation of these complexes, which can also contaminate agricultural products and harm human health (Gros et al., 2021). AMCs are closely linked to the emergence of antibiotic resistance genes (ARGs), which contribute to the spread of antibiotic-resistant infections and pose a serious global health threat (Zhao et al., 2023). Resistant bacteria can transfer these genes to healthy organisms, reducing the effectiveness of antibiotics and exacerbating the problem of resistance, especially in low-income and developing nations, where resources are limited (Zainab et al., 2020). Addressing this challenge requires the development of new antibiotics, that involve significant optimization efforts and investment (Turek et al., 2024). A reliable approach for evaluating potential ecological risks of AMCs across various ecosystems is to calculate the Risk Quotient (RQ).

$$RQ = \frac{MEC(\text{Measured Environmental Concentration})}{PNEC(\text{Predicted No - Effect Concentration})}$$

Laboratory experiments and field data provide MEC, while toxicity tests, such as EC₅₀, conducted on a sensitive organism are used to derive the PNEC after adjusting for safety parameters. If RQ ≤ 1, the risk is negligible. However, if RQ > 1, the risk is significant and requires immediate action.

6. Mitigation and management strategies

The presence of AMCs in wastewater is a significant environmental concern because of their low biodegradability and high chemical stability, which pose threats to both the environment and human health

(Jabbar et al., 2024). This highlights the urgent need for effective remediation technologies to mitigate the impacts of these pollutants. Given the complexity of removing both metal and antibiotic residues, integrated treatment approach is necessary. These methods aim to reduce the metal-antibiotic complexation phenomenon, which often results from the filtering of agricultural and industrial pollutants into water reservoirs. Each remediation method has advantages and challenges, which must be carefully considered when determining the most effective treatment approach.

6.1. Remediation technologies for AMCs

Environmental heterogeneity, variations in pH and temperature, and the presence of competing contaminants present challenges in the implementation of remediation technologies beyond controlled laboratory settings. While single methods, such as phytoremediation, adsorption, or bioremediation, may be effective in the laboratory, integrated strategies that combine physical, chemical, and biological approaches, supported by intensive monitoring, are more likely to succeed in the field.

6.1.1. Adsorption

Adsorption is an effective method for removing AMCs from wastewater because of its high efficiency, cost-effectiveness, and versatility for handling a wide range of contaminants (Yao et al., 2020). Common adsorbents include activated carbon, nanoparticles, and biochar, which are preferred because of their high surface areas and customizable surface properties (Li et al., 2024). However, the co-presence of metal ions and antibiotics can influence adsorption behavior. Metals may compete with antibiotics for active adsorption sites or reduce their interactions with the adsorbent, thereby decreasing the adsorption capacity for organic contaminants (Chen et al., 2024c). Conversely, metals adsorbed first can act as bridges, linking organic molecules to the adsorbent and enhancing antibiotic adsorption (Khurana et al., 2021b). This "decomplexing-bridging" mechanism, which involves the breakdown of AMCs followed by interactions between metals and the adsorbent, is pH-dependent (Zhang et al., 2021). Additionally, inorganic electrolytes and organic matter in the wastewater matrix can negatively impact the adsorption efficiency by competing for active sites or weakening the interactions between metals, antibiotics, and the adsorbent.

6.1.2. Coagulation-flocculation processes

Coagulation-flocculation processes involve the formation of flocs that capture contaminants, which are then removed through sedimentation or filtration. This method is particularly effective for treating wastewater that contains mixed pollutants.

6.1.3. Membrane filtration

Methods such as reverse osmosis, nanofiltration, and ultrafiltration physically separate antibiotics from water, based on their size and charge. These processes are highly effective, but can be costly and require regular maintenance.

6.1.4. Chemical precipitation

Chemical precipitation involves adding substances, such as lime or ferric chloride, to wastewater, forming precipitates that can be easily removed through sedimentation or filtration (Xu et al., 2024). This method is effective for removing metal ions, but may have limited efficacy for AMCs.

6.1.5. Ion exchange

Ion-exchange resins selectively remove antibiotics from water by swapping with similarly charged ions on the resin surface (Luo et al., 2024). This method is effective for specific contaminants, but may require frequent regeneration of resins.

6.1.6. Phytoremediation

Phytoremediation uses plants, soil, and associated microbial assemblages to reduce pollutants in wastewater. It is often employed in engineered treatment systems and constructed wetlands, offering a low-cost, flexible, and eco-friendly in-situ technique for removing heavy metals and co-occurring antibiotics (Guo et al., 2019).

6.1.7. Advanced oxidation processes (AOPs)

AOPs degrade persistent organic pollutants via reactive oxygen species (ROS) such as hydroxyl and superoxide radicals (Pignatello et al., 2006). These processes offer non-selective oxidation and rapid reactions. These include UV-based, ozone-based, physical, electrochemical, and heterogeneous photocatalysis methods. Semiconductor photocatalysis utilizes visible light to efficiently degrade pollutants, making it a sustainable and cost-effective solution for environmental remediation (Jabbar et al., 2024). Fig. 6 highlights the various AOP techniques employed for the effective degradation and removal of AMC from contaminated environments.

6.2. Strategies for addressing aquifer pollution from antibiotic-metal complexes

Under the Clean Water Act (CWA) and Safe Drinking Water Act (SDWA), the EPA has set guidelines to limit the concentrations of heavy metals and pharmaceutical residues, including antibiotics, in water bodies to protect human health and aquatic life. International organizations such as the United Nations Environment Programme (UNEP) and the World Health Organization (WHO) emphasize the need for global cooperation in research, policy-making, and the development of new technologies to control complex pollutants in the environment. However, comprehensive international regulations specifically targeting antibiotic-metal complex (AMC) pollutants are still lacking, as research in this area is ongoing and not yet mature enough to inform policy development.

An integrated strategy is crucial for addressing the AMC pollution in aquifers. This approach should incorporate advanced wastewater treatment technologies such as ion exchange, activated carbon filtration, and efficient oxidation procedures (Pei et al., 2019). Long-term solutions include preventive measures, such as improved farming practices and proper disposal of pharmaceutical waste, to prevent further contamination. Policymakers should enforce stringent regulations and implement robust zoning and land-use planning strategies to govern potentially hazardous activities (Jain, 2023). Public awareness of the hazards associated with AMC pollution and the importance of remediation is essential. Collaboration among governmental agencies, non-governmental organizations, industries, and the scientific community is vital for effectively protecting aquifers and groundwater.

7. Future research directions

7.1. Understanding chemical and biological mechanisms

A multifaceted research approach is essential for tackling the challenges posed by AMCs. The investigation began with an in-depth study of the chemical and biological mechanisms that govern the formation and degradation of AMCs in various environmental settings. This involves analyzing the interactions between antibiotics and metals under different conditions, such as pH, temperature, and the presence of other ions, to understand how these factors influence the stability and transformation of AMCs.

7.2. Development of transport models

The next crucial step involves developing and refining transport models to predict the movement and fate of AMCs in aquifer systems. These models must incorporate complex processes, such as adsorption to soil particles, desorption back into the water, diffusion through different media, and chemical reactions that can alter the composition and toxicity of AMCs over time. Model validation is essential to ensure accuracy and practical applicability. This can be achieved through tracer studies, which track the movement of these complexes in controlled environments, and field experiments, which observe their behavior in real-world scenarios. A detailed understanding will not only improve the predictability of AMC behavior but also inform strategies for mitigating their potential risks to ecosystems and human health.

7.3. Ecotoxicological impact assessment

In conjunction with chemical and transport studies, ecotoxicological research is crucial for understanding the broader environmental impact of AMCs. These studies should examine how the presence of both metals and antibiotics affects aquatic organisms, considering both the acute and chronic effects. Special attention should be given to the potential for bioaccumulation and biomagnification, where the concentrations of AMCs increase throughout the food chain, posing risks to higher trophic levels, including humans.

7.4. Advanced remediation technologies

There is an urgent need to explore and develop advanced remediation technologies for the efficient removal of AMCs from contaminated water sources. Techniques such as advanced oxidation processes, nanomaterials designed for the adsorption or catalysis of AMC degradation, and bioremediation approaches using microorganisms to detoxify environments must be assessed for their effectiveness and sustainability. This evaluation should also consider the environmental and economic impacts to ensure practical implementation.

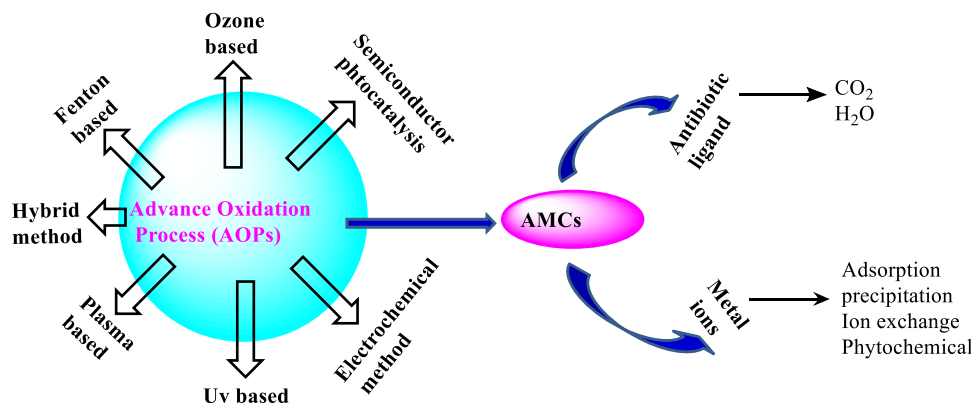


Fig. 6. Advanced Oxidation Processes (AOPs) used in the remediation of antibiotic-metal complexes.

7.5. Policy development and public awareness

Collaboration with policymakers is essential to establish comprehensive regulations and guidelines for AMC pollution control. Ensuring that wastewater treatment and disposal practices minimize the release of these harmful complexes into the environment is crucial. Additionally, public awareness campaigns are vital to educate communities about the sources and dangers of AMC pollution, promoting responsible behaviors such as proper disposal of pharmaceuticals and support for environmental monitoring initiatives. A holistic approach that integrates chemical, ecological, and policy studies with innovative technological solutions is key to addressing AMC challenges. By combining scientific research with public engagement and regulatory frameworks, we can mitigate the risks associated with AMCs and protect both the environment and public health. In order to mitigate AMC pollution globally, it is imperative to mandate antibiotic use reporting across the healthcare, agricultural, and veterinary sectors, as well as enforcing strict emissions standards for industries that discharge heavy metals. Public awareness campaigns should focus on educating stakeholders about AMC risks and encouraging proper disposal of antibiotics. In addition to recycling metals and implementing antibiotic take-back programs, circular economic practices can reduce environmental inputs. In addition, standardized monitoring programs for AMC levels in the environment should be established and integrated into current antibiotic surveillance systems to improve tracking and management. Advances in technology for advanced wastewater treatment plants are urgently needed globally, enabling them to be shared universally.

8. Conclusion

Metal contamination of aquatic environments and the widespread use of antibiotics pose significant ecological and public health risks. Transition metals such as copper, iron, zinc, nickel, and cobalt readily form stable complexes with antibiotics, owing to their reactivity and coordination properties. Metal-based medications, including antibiotic-metal complexes, have shown potential in treating severe diseases, such as cancer and neurodegenerative disorders, by enhancing antibiotic activity, increasing bioavailability, and allowing for reduced dosages. However, the stability of these complexes is influenced by the chemical structure of the antibiotic and nuclear charge of the metal ion. Despite their therapeutic benefits, AMCs can be more toxic than their parent compounds, contributing to bacterial toxicity and promoting the development of antibiotic-resistance genes (ARGs). Pollution from AMCs in aquifers poses severe health risks including gastrointestinal issues, organ damage, and cancer. These complexes disrupt ecosystems, reduce biodiversity, and alter species composition, thereby affecting the higher trophic levels. The bioaccumulation of AMCs in the food chain can contaminate agricultural products and affect human health. Furthermore, the emergence of ARGs complicates global resistance issues and increases the complexity and cost of remediation. Industries, such as fertilizer production, textiles, and paper manufacturing, release harmful pollutants, including heavy metals and antibiotics, into water bodies. Effective remediation technologies are crucial for addressing this contamination. Integrated treatment approaches, which combine methods such as phytoremediation, adsorption, membrane filtration, ion exchange, and advanced oxidation processes, offer promising solutions for removing complex contaminants. These approaches are essential to safeguard water resources and protect public health.

CRediT authorship contribution statement

Ajaya Bhattarai: Writing – review & editing. **Narendra Kumar Chaudhary:** Writing – review & editing, Supervision, Software. **Dipak Baral:** Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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